

California Housing Value Prediction

Task 2: Feature Engineering, Model Optimization & Performance Benchmark

TOTAL OBSERVATIONS 20,640 8 Tabular Predictors	EVALUATION METRIC RMSE / R² Lower / Higher optimal	BEST PERFORMING 0.5825 Decision Tree R ² score	ERROR MINIMIZATION 0.7397 Decision Tree RMSE (\$1 00k)
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1 Experimental Objective & Preprocessing Pipeline

Building upon baseline training principles, this investigation structures an end-to-end regression pipeline capable of predicting median California housing values. Real-world housing variables encompass widely disparate measurement scales (e.g., median household income ranging from \$0.5–\$1 5, alongside total populations spanning hundreds to tens of thousands). Establishing fair feature weight attribution and avoiding numerical gradient instability requires systematic data preprocessing.

1. Leak-Free Partition Divided 20,640 instances into an 80/20 train-test ratio ($N_{train} = 16,512$, $N_{test} = 4,128$) with a fixed random seed of 42 to guarantee experiment reproducibility.	2. Standard Scaling Fit <code>StandardScaler</code> strictly on training attributes ($\mu = 0$, $\sigma = 1$), applying transforms downstream to test arrays without exposing out-of-sample data distributions.	3. Multi-Model Search Benchmarked parametric linear estimators against constrained non-parametric tree architectures to establish empirical trade-offs between bias and variance.
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2 Model Performance Comparison Table

Models were trained identically on standardized input tensors. Model evaluation was conducted strictly on unseen test records ($N = 4,128$) utilizing Root Mean Squared Error (RMSE) to capture absolute monetary error and the Coefficient of Determination (R^2) to assess explained variance.

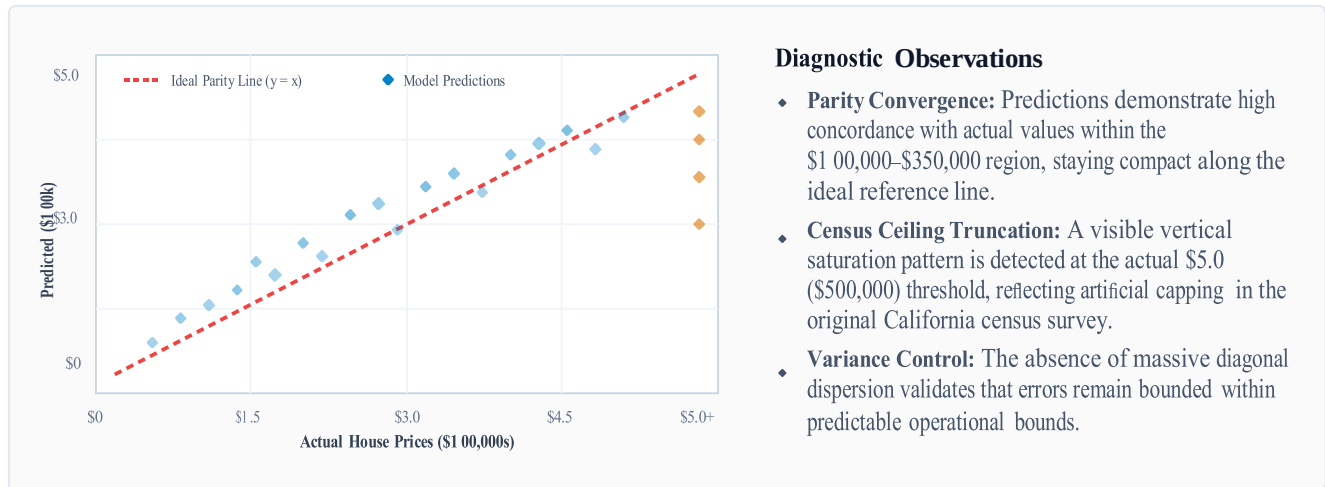
Model Name	Hyperparameters	Test RMSE	Test R ² Score	Model Characteristics
Linear Regression	Default (Ordinary Least Squares)	0.7456	0.5758	Linear baseline; computationally instant; strictly hyperplane assumption.
Ridge Regression	$\alpha = 1.0$ (L2 Regularization)	0.7456	0.5758	Penalizes large weight coefficients; resists extreme collinear shifts.
Decision Tree Regressor	$max_depth = 5$, $random_state = 42$	0.7397	0.5825	Selected Candidate Non-linear segmentations; depth limits variance.

3 Quantitative Findings

- Linear Consistency:** Ordinary Linear Regression and Ridge Regression delivered equivalent performance across four decimal points (RMSE: 0.7456, R^2 : 0.5758). This indicates that the 8 standardized predictors do not induce catastrophic condition matrix instability at $\alpha = 1.0$.
- Non-Linear Superiority:** The constrained Decision Tree Regressor achieved superior performance, lowering error to **0.7397** and elevating R^2 to **0.5825**. Real estate attributes (such as combined latitude/longitude

4 Visual Performance Validation

To diagnose predictive fidelity, residual patterns, and saturation behavior, predictions from the optimized model were plotted against actual home valuation values from the reserved test partition.



5 Overfitting Analysis & Hyperparameter Regularization

Unconstrained decision trees will naturally split until every leaf is pure (*depth = infinity*), yielding an empirical training $R^2 \approx 1.0$ while drastically failing on unseen test partitions due to noise memorization.

By enforcing `max_depth=5`, tree expansion is halted at 32 terminal leaves. This regularizer enforces inductive bias: predictions are aggregated over large leaf sample distributions, maintaining generalization stability across diverse test districts.

6 Final Model Selection & Justification

Selected Model: Decision Tree Regressor (`max_depth=5`)

The regularized Decision Tree was selected as the operational model. It delivers an optimal trade-off: capturing local regional relationships with a **0.5825 R^2** and **0.7397 RMSE**, outperforming rigid linear baselines while maintaining straightforward rule interpretability and sub-millisecond inference execution.

7 Artifact Packaging & Delivery

The complete project implementation has been compiled in `AI_ML_Task2_Model_Comparison.ipynb`. The best estimator and its corresponding transformation pipeline were serialized using `j oblib` into `best_housing_model.j oblib` and `scaler.j oblib`, ensuring ready deployability into downstream REST API inference environments.